EASA COLLEGE OF ENGINEERING AND TECHNOLOGY
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TEXT ANALYTICS

TERM PROJECT REPORT

Role of Information Extraction in Building Knowledge Graphs: A Case Study on Google Knowledge Graph

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Abstract

Knowledge graphs (KGs) store facts as entities connected by relations and power modern search, question answering and recommendation systems. Most real-world knowledge is stored in unstructured text, so Information Extraction (IE) is the key step that converts text into structured knowledge. This project builds a small end-to-end pipeline in Python using spaCy, NetworkX and PyVis that performs Named Entity Recognition, relation extraction, triple generation and graph visualisation. The pipeline is then compared with the Google Knowledge Graph, a large-scale real-world KG, to study the role, strengths and limitations of information extraction in building knowledge graphs.

1. Introduction

A knowledge graph represents information as a network in which nodes are entities (people, organisations, places) and edges are relationships between them. Each fact is stored as a triple of the form (subject, relation, object), for example (Larry Page, founded, Google). Google introduced its Knowledge Graph in 2012 with the idea of understanding "things, not strings", which means that a search engine should understand real-world entities instead of only matching keywords.

The text on the web is unstructured, so it cannot be placed directly into a graph. Information Extraction is the group of Text Analytics techniques that identifies entities and relations in raw text and converts them into structured triples. Without IE, large knowledge graphs could not be built or kept up to date.

1.1 Objectives

- To explain the role of information extraction in building knowledge graphs.
- To implement a working pipeline: text, NER, relation extraction, triples and graph.
- To study the Google Knowledge Graph as a real-world case study.
- To compare the small pipeline with Google KG and discuss limitations.

2. Information Extraction and Knowledge Graphs

2.1 Main IE tasks

- Named Entity Recognition (NER): finding and labelling entities such as PERSON, ORG, GPE and DATE.
- Entity Linking / Disambiguation: mapping a mention such as "Google" to the correct unique entity in the KG.
- Relation Extraction: identifying how two entities are related, for example founded, acquired, CEO of.
- Triple generation: writing each fact as (subject, relation, object).

- Knowledge fusion: merging facts from many sources and removing duplicates and conflicts.

2.2 How IE feeds a knowledge graph

The flow is: raw text, then NER, then entity linking, then relation extraction, then triples, then the knowledge graph. The quality of the graph depends directly on the accuracy of each earlier stage. An error in NER or in relation extraction becomes a wrong or missing edge in the graph.

3. Methodology

3.1 Tools and environment

Tool	Purpose
Python 3 / Google Colab	Development environment
spaCy (en_core_web_sm)	Tokenisation, NER, dependency parsing
NetworkX	Building the directed graph
Matplotlib	Static graph visualisation
PyVis	Interactive graph visualisation

3.2 Input text

A short text about Google was used as input:

```
Google was founded by Larry Page and Sergey Brin in 1998.
Google is headquartered in Mountain View, California.
Sundar Pichai is the CEO of Google.
Google acquired YouTube in 2006.
Google launched the Knowledge Graph in 2012.
```

3.3 Step 1: Named Entity Recognition

The spaCy model en_core_web_sm was loaded and applied to the text. The model identified the entities listed in Section 4.1.

3.4 Step 2: Relation extraction

Two approaches were used. First, a curated list of triples was prepared from the sentences. Second, an automatic rule-based extractor was written using spaCy dependency parsing: for every verb, the subject (nsubj / nsubjpass) on its left and the object (dobj / attr) on its right were taken, and the verb lemma was used as the relation.

3.5 Step 3: Graph construction and visualisation

A directed graph was built with NetworkX, where each triple adds an edge from subject to object with the relation as the edge label. The graph was drawn with Matplotlib and also as an interactive, draggable graph with PyVis.

4. Results

4.1 Named entities found

Entity	Label	Meaning
Google, YouTube, the Knowledge Graph	ORG	Organisation
Larry Page, Sergey Brin, Sundar Pichai	PERSON	Person
Mountain View, California	GPE	Geo-political entity
1998, 2006, 2012	DATE	Date

4.2 Triples (curated)

Subject	Relation	Object
Larry Page	founded	Google
Sergey Brin	founded	Google
Google	headquartered in	Mountain View
Mountain View	located in	California
Sundar Pichai	CEO of	Google
Google	acquired	YouTube
Google	launched	Knowledge Graph

4.3 Triples (automatic extraction)

Subject	Relation	Object
Google	acquire	YouTube
Google	launch	Graph

The automatic extractor found only 2 of the 7 facts, and one of them is incomplete because "Knowledge Graph" was reduced to "Graph".

4.4 Knowledge graph output

The final graph contains 8 nodes (Google, Larry Page, Sergey Brin, Sundar Pichai, YouTube, Knowledge Graph, Mountain View and California) connected by labelled directed edges. Google is the central node, linked to its founders, CEO, headquarters, an acquisition and a product. The static graph (Matplotlib) and the interactive graph (PyVis) screenshots are attached in the output screenshots PDF and in the GitHub repository.

5. Case Study: Google Knowledge Graph

5.1 Overview

The Google Knowledge Graph was launched in 2012. It is a large database of entities and the relationships between them. It is used to show Knowledge Panels, direct answers and related entities in Google Search and in Google Assistant.

5.2 Role of information extraction

- Sources: Wikipedia, Wikidata, the CIA World Factbook, licensed data, structured markup on web pages, and text from web documents.
- NER and entity linking identify entities in web text and connect each mention to a single unique entity.
- Relation extraction at web scale, using machine learning, finds new facts from text.
- Knowledge fusion combines facts from many sources and uses confidence scores to resolve conflicts.
- Continuous updating keeps the graph current as new text is published.

5.3 Our pipeline compared with Google KG

Aspect	Our pipeline	Google Knowledge Graph
Scale	1 short text, 7 triples	Billions of facts
Data sources	Single plain text	Web, Wikipedia, Wikidata, structured data
Extraction method	spaCy NER + rule-based dependency rules	Machine learning and structured data
Entity linking	Not implemented	Core component
Fusion / validation	None	Multi-source fusion with confidence scores
Output	NetworkX / PyVis graph	Knowledge Panels in Search

6. Limitations and Discussion

- Multi-word entities can be split, for example "Knowledge Graph" became "Graph" in automatic extraction.
- Rule-based extraction misses many facts, since it only handles simple subject-verb-object sentences. Relations such as "CEO of" and "headquartered in" were missed.
- No entity linking, so ambiguous names (Google the company vs. other uses) cannot be resolved.
- Small en_core_web_sm model has limited accuracy; for example, "the Knowledge Graph" was labelled ORG rather than a product.

- Scale: the toy pipeline works on one text, whereas real KGs need distributed, continuously updated pipelines.

These limitations show why large systems such as Google KG combine machine-learning-based extraction, entity linking, structured data and fusion, instead of relying on simple rules.

6.1 Future work

- Use a larger model (en_core_web_trf) or a transformer-based relation extraction model.
- Add entity linking against Wikidata.
- Extract triples from many Wikipedia pages and compare them with Wikidata facts.

7. Conclusion

This project showed that information extraction is the foundation of knowledge graph construction: NER finds the entities, relation extraction finds the connections, and triples turn them into a graph. The implemented pipeline successfully converted a short text into a knowledge graph with 7 facts, and the comparison with Google Knowledge Graph showed that real systems need entity linking, machine learning at scale, multiple sources and knowledge fusion. The analysis of extraction errors also gives a clear view of why high-quality extraction is the hardest and most important step in building a knowledge graph.

8. Project Links

- GitHub repository: https://github.com/rakavirakesh-dev/kg-information-extraction
- Deliverables: Jupyter Notebook (.ipynb), report (PDF), output screenshots (PDF), demo video (MP4).

9. References

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- PyVis documentation: https://pyvis.readthedocs.io
- Wikidata: https://www.wikidata.org